



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PALANI VELAYUDAM TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

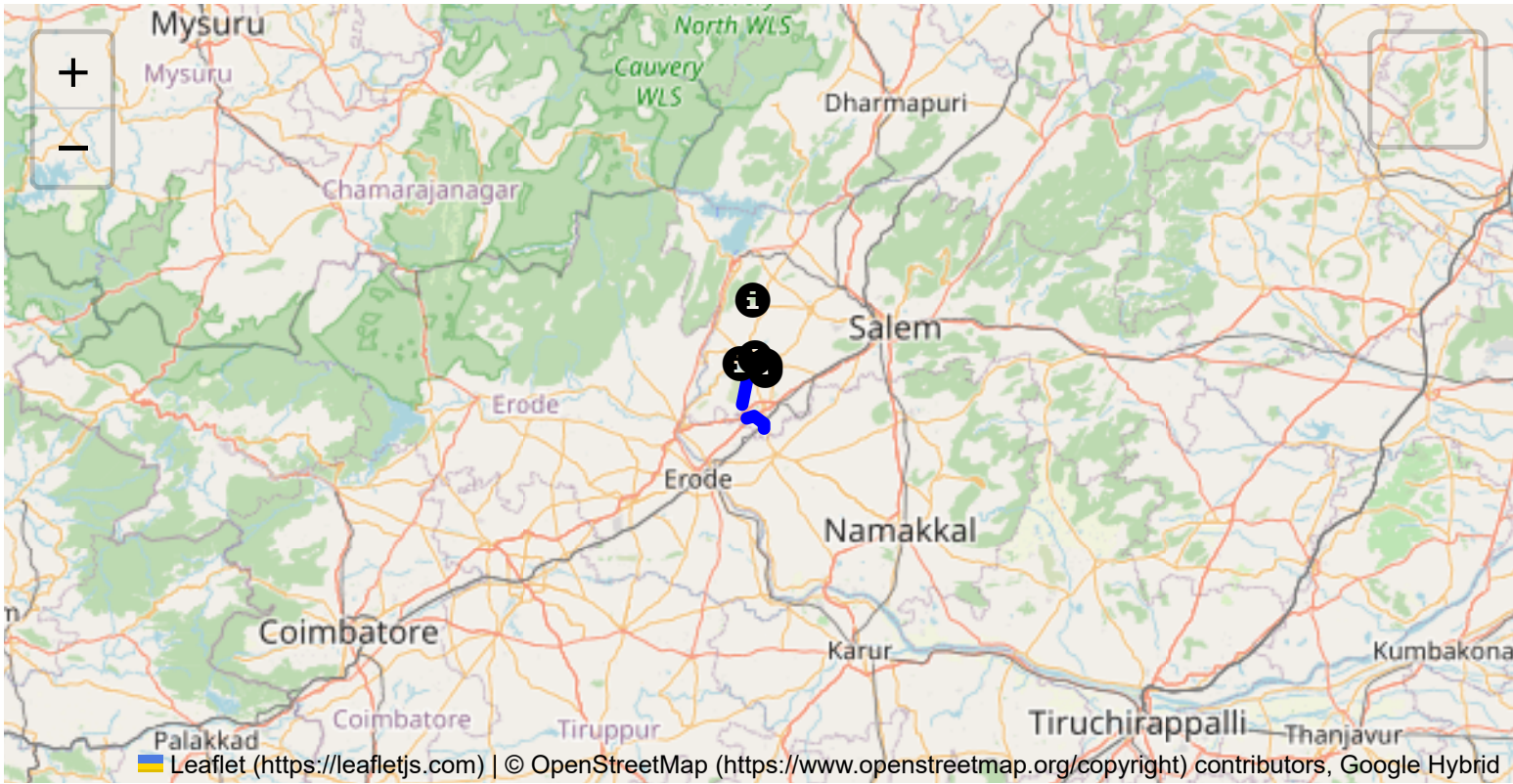
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

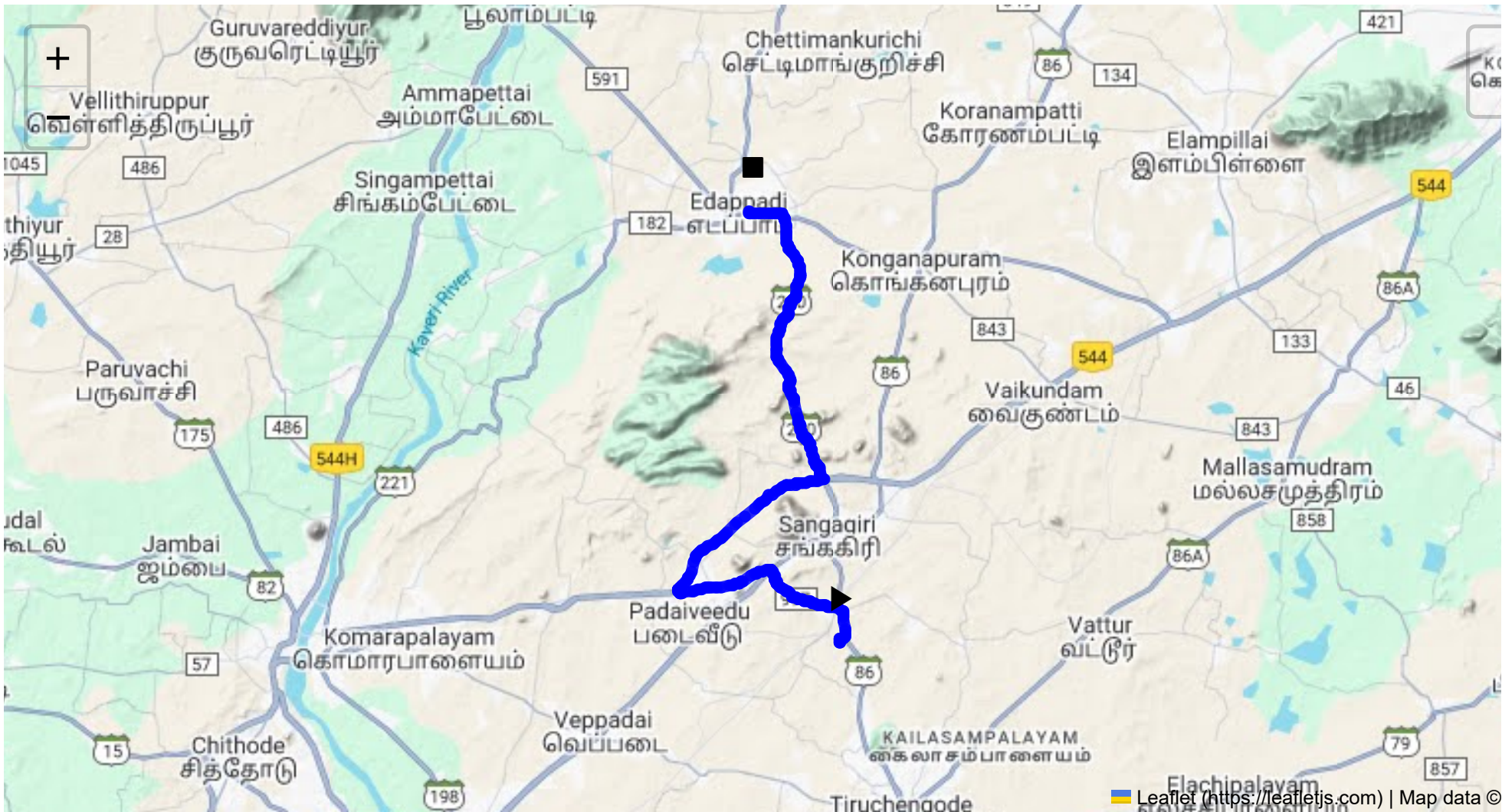
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 28.15 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.585389, 77.84204)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route stretches approximately 28.15 kilometers, running from Sangagiri through key waypoints to Edappadi. It primarily follows regional roads, such as the Sankari RS - Erode Privu Road and Bhavani Main Road. The journey directs through semi-urban areas intertwined with rural patches, offering a mix of two-lane and sometimes wider roads suitable for heavy vehicles.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences three main seasons: summer, monsoon, and winter. This region can be hot and humid, especially during March to May. The monsoon period from June to September may result in heavy rainfall, leading to slippery roads and possible flooding in under-maintained areas. The post-monsoon season can still have residual showers.

3. Analysis of Traffic Patterns

Traffic congestion is more pronounced in towns along the Bhavani Main Road, notably near marketplaces and school zones. Peak hours align with work commutes (8-10 AM, 5-7 PM) and school timings. Sundays and festival days might observe higher traffic due to local events and shopping.

4. Assessment of Road Quality and Infrastructure

Road quality varies; while some stretches are well-maintained asphalt, others may have potholes or uneven surfaces. The Bhavani Main Road could become narrow in certain sections, and drivers need to be cautious about unmarked speed breakers and pedestrian crossings.

5. Suggestions for Alternative Routes for Emergencies

An alternate route could be bypassing major towns via less congested rural roads, but this requires local knowledge. Check current conditions using GPS systems for real-time detours in case of traffic or accidents.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transportation of hazardous materials requires adherence to Tamil Nadu Pollution Control Board guidelines and the Motor Vehicles Act, 1988. Proper permits, markings, and adherence to specified routes may be enforced, and driving during night hours might be restricted.

7. Overview of Historical Incidents

While specific incident data is region-dependent, typical risks include vehicular accidents at busy intersections or due to road conditions. Consider liaising with local authorities for detailed historical data.

8. Environmental Considerations and Sensitive Areas

The route does not pass through any major environmental protection zones, but awareness of local flora and fauna crossings is crucial. Respect for farmlands and water bodies is imperative and waste disposal should comply with regulations.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable, especially in towns and along major roads. However, intermittent dead zones might occur in rural sections. It's advisable to have backup communication methods such as satellite phones or company radio systems.

10. Estimated Emergency Response Times

Emergency services in towns like Sankari are relatively quick, ranging from 10 to 20 minutes. In remote stretches, response times may increase significantly, exceeding 30 minutes due to accessibility challenges.

11. Overall Summary of Risk Assessment

The journey presents moderate risk due to variable road conditions and weather impacts. Congestion in urban centers necessitates vigilance, while road quality may affect vehicular safety. Emergency preparedness, awareness of local regulations, and environmental considerations support safe operations. Equip vehicles with emergency kits and ensure all permits are current. Regular communication with logistics coordinators and possibly a local guide may optimize safety during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45348, 77.85706	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	Medium	11.49249, 77.85820	30 KM/Hr
11	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
12	Turn	Medium	11.58296, 77.85419	30 KM/Hr
13	Turn	High	11.58516, 77.84187	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	KRS Hospital and Diabetic Centre	11.5848828, 77.8449664	30 km/h	Medium
2	hospital	Kannan Hospital	11.5854802, 77.8425936	30 km/h	Medium
3	hospital	Aravind Hospital	11.5854615, 77.8424165	30 km/h	Medium
4	hospital	Muthu Eyecare Hospital	11.584736, 77.8393794	30 km/h	Medium
5	clinic	Uma Clinic	11.58398, 77.83977	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45348, 77.85706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49249, 77.85820



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58296, 77.85419



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58516, 77.84187

